

Partially inelastic surface — Solution

X22 (2022) — English translation

A1

0.10 pts

Express $\dot{v}_x$ in terms of F_x and m .

From the theorem on the motion of the center of mass

$$F_x = m\dot{v}_x$$

hence

$$\dot{v}_x = \frac{F_x}{m}$$

Answer: \textbf{Answer:}

$$\dot{v}_x = \frac{F_x}{m}$$

A2

0.40 pts

Express $\dot{\omega}$ in terms of F_x , N , L , φ and m .

From the equation of dynamics of rotational motion relative to the center of mass

$$I_O\dot{\omega} = \frac{mL^2\dot{\omega}}{3} = NL \sin \varphi - F_x L \cos \varphi$$

hence

$$\dot{\omega} = \frac{3}{mL} (N \sin \varphi - F_x \cos \varphi)$$

Answer: \textbf{Answer:}

$$\dot{\omega} = \frac{3}{mL} (N \sin \varphi - F_x \cos \varphi)$$

A3

0.10 pts

Express v_{Ax} in terms of ω , v_x , φ and L .

From the theorem on the addition of velocities

$$\vec{v}_A = \vec{v}_O + \vec{v}_{O \rightarrow A}$$

hence

$$v_{Ax} = v_x - \omega L \cos \varphi$$

Answer: \textbf{Answer:}

$$v_{Ax} = v_x - \omega L \cos \varphi$$

A4

0.40 pts

Prove that $v_{Ax} \leq 0$ for any values of μ and φ .